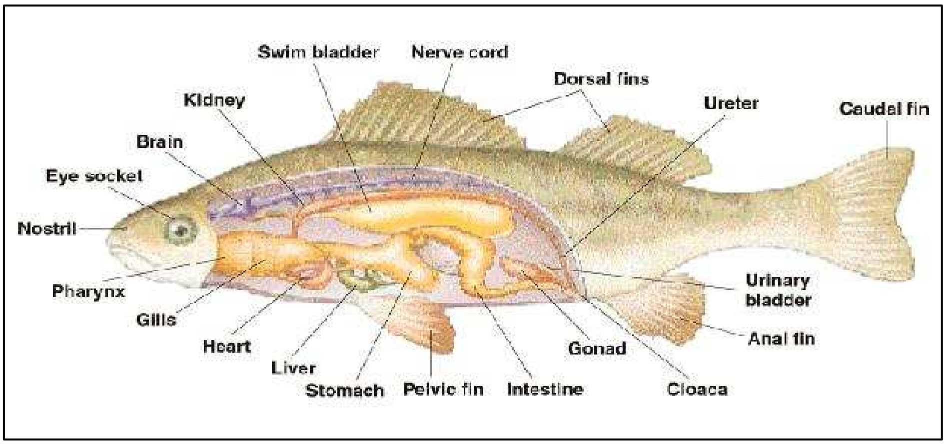
Fish dissection guide

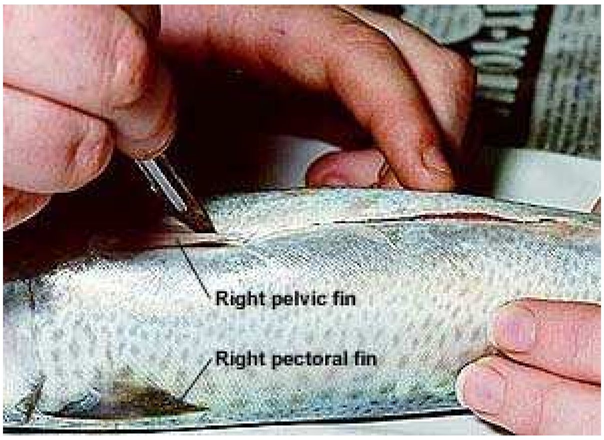
Internal Anatomy:



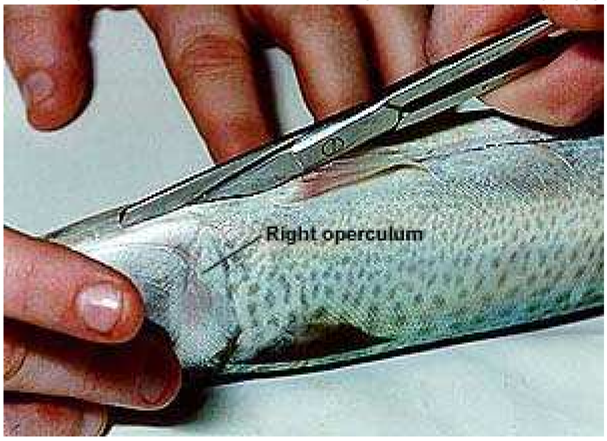
1. Incision at anus. Begin by inserting a fine scalpel blade into the anus (also called the vent) of the fish. The anus is located just anterior to (in front of) the anal fin, on the ventral (lower) side of the fish in most fishes.

2. Cutting anteriorly. The incision is then extended anteriorly along the fish's belly towards the head.

3. Cut between pelvic fins. The incision passes anteriorly between the pelvic (ventral) fins. Depending on the type of fish, these paired fins are used to stabilise the fish when swimming and also for braking. The pelvic fins are supported by the bones of the pelvic girdle which are anchored in the belly muscles.

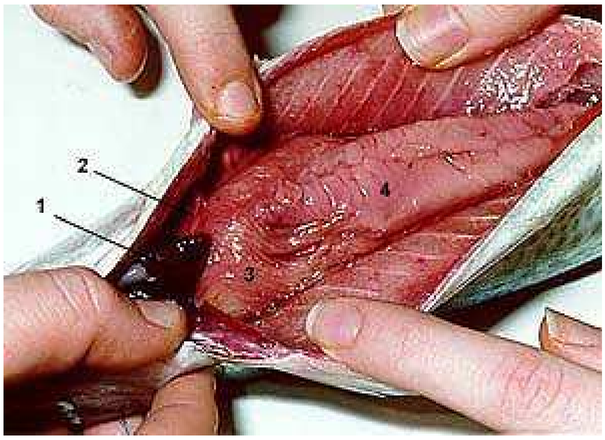


4. Cut along isthmus. Use scissors to cut anteriorly through the bones attached to the pelvic fins. Cut forward along the narrow, fleshy space beneath the head and between the gill covers. The gill covers (also known as operculae) are flaps which lie along both sides of the head and protect the underlying gills.

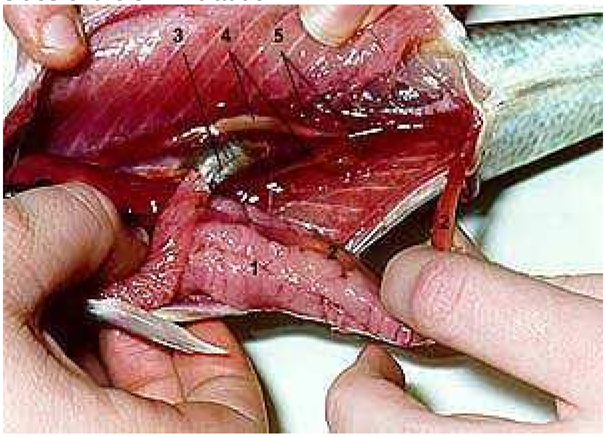


5. Body cavity. Pull apart the two walls of the body cavity and expose the internal organs (see next image for names). The neat incision now runs from the anus forward between the two pelvic fins and along the isthmus.

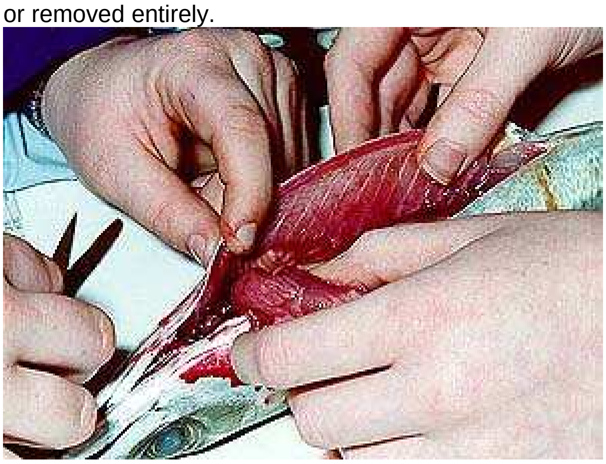
6. Internal organs. Some of the ventrally located internal organs: 1 heart, 2 Liver, 3 Pyloric caecae, 4 adipose (fatty) tissue



7. Pull aside gut. Here the adipose tissue (1) and gut (2) are pulled aside to expose the swim bladder (3), gonads (4) and kidneys (5). As a general rule, carnivorous fishes have short guts. Herbivorous fishes have much longer guts. The gonads and kidneys are paired. One of each can be seen on both sides of the swim bladder.

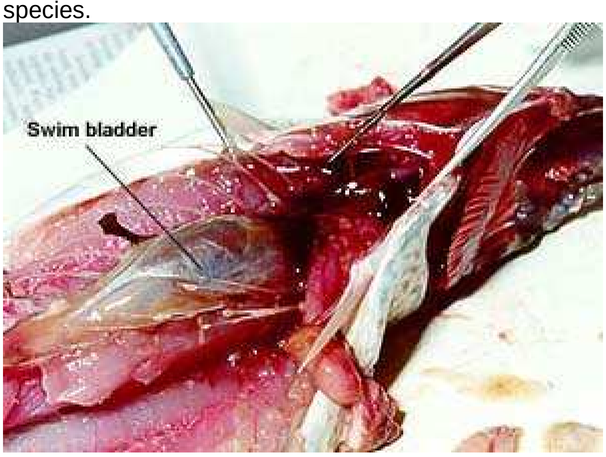


8. Cut posterior end of gut. The gut is severed at the posterior end of the body cavity, near the anus. The gut and other organs attached to it are pulled forward out of the way, or removed entirely.

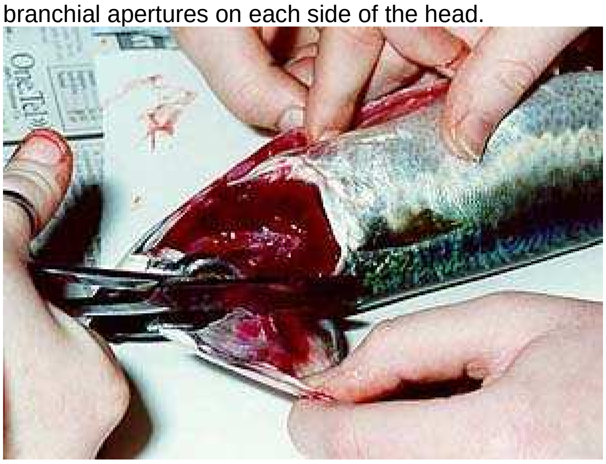


9. Pull gut forward. Pulling the gut forward exposes the swim bladder (1), gonads (2) and kidneys (3) in position dorsally (at the top) in the body cavity. A larger portion of the liver is now visible (4). The kidneys are paired organs located in the body cavity ventral to (below) the vertebral column. They are one of the organs involved in excretion and regulation of the water balance within the fish.

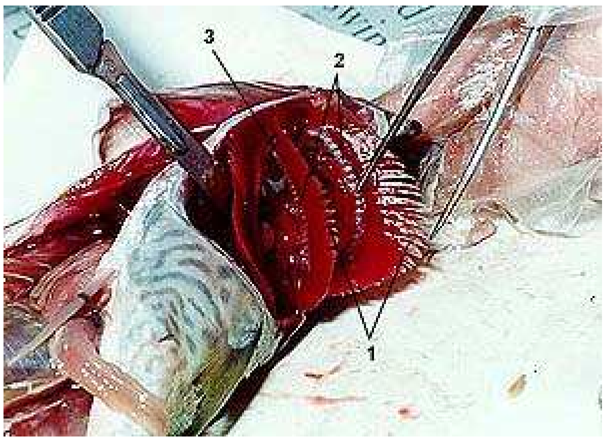
10. Swim bladder exposed. The other organs have been removed to expose the swim bladder at the top of the body cavity. The swim bladder (also called the gas bladder or air bladder) is a flexible-walled, gas-filled sac located in the dorsal portion of body cavity. This organ controls the fish's buoyancy and is used for hearing in some species.



11. Cutting operculum. Here, the right gill cover (operculum) is being removed to expose the underlying gills. Most bony fish have the characteristic of having a single opening behind each operculum (the branchial aperture). Water passes in through the mouth, over the gills and out through the branchial aperture. In contrast, the sharks and rays have five to seven branchial apertures on each side of the head.



12. Gills exposed. Most gills consist of gill filaments (1), gill rakers (2) and gill arches (3). Gills of fishes are the sites where oxygen is absorbed and carbon dioxide is removed. In addition, the gills are responsible to a varying degree for regulation of the levels of various ions and the pH of the blood. Gill rakers are bony or cartilaginous projections that point forward and inward from the gill arches. They aid in the fish's feeding.



Fish anatomy – Organ Function

<u>The heart</u>. The circulatory system in fishes is a single circuit, with blood flowing from the heart to the gills and then to the rest of the body. The heart is located a little behind and below the gills. The typical fish heart has four chambers, however unlike mammals, blood moves through all four in sequence. The heart of slow moving fishes is comparatively small, whereas active swimming species have large hearts.

<u>The liver.</u> The liver has many digestive and storage functions. One of these is the production of bile, a solution which emulsifies fats and may assist in changing the acidic conditions of the stomach into the neutral pH of the intestine. The liver is also responsible in some species for the storage of fats, blood sugar, and vitamins A and D. Before it was possible to synthetically create vitamins A and D, sharks were caught for their livers which have high concentrations of these vitamins.

<u>Pyloric caecae</u>. Pyloric caecae (singular caecum) are finger-like pouches connected with the alimentary canal (the gut). They are attached to the pylorus, the section of the intestinal tract immediately following the stomach. They range in number from three in a type of scorpion fish to thousands in tuna. Pyloric caecae may have a digestive and/or absorptive function. The enzyme lactase has been found in the pyloric caecae of some fishes such as trout.

<u>Gonads.</u> The sexes of fishes are usually separate. Males usually have paired testes that produce sperm, and females usually have paired ovaries that produce eggs. When paired, the gonads lie on either side of the swim bladder. The method by which the eggs and sperm meet and thus fertilisation occurs varies widely among fishes. Many species are broadcast spawners, shedding their eggs and sperm into the water to fertilise external to the body. Other species such as sharks and rays have internal fertilisation where the sperm are released into the body of the female. Many variations exist, including the seahorse, in which the female deposits her eggs into the pouch of the male where they are fertilised. The hagfishes and lampreys have a single ovary or testis. Sperm and eggs are shed into the body cavity and out through a urogenital papilla.

<u>The Kidneys</u>. These are one of the organs involved in excretion and regulation of the water balance within the fish. Freshwater and marine fishes are faced with different problems with regard to regulating the concentration of salts within the body and as such, their kidneys differ considerably in structure. Freshwater fishes have larger kidneys than marine fishes. They have a higher concentration of salts in the body tissues than the surrounding water. Conversely marine fishes have a lower concentration of salts in the body tissues than the surrounding water. The kidneys of freshwater fishes remove water and re-absorb salts and sugars. They produce large amount of very dilute urine. This helps the fish avoid becoming "waterlogged" from the large amounts of water diffusing into the fish. The kidneys of marine fishes however conserve water. Marine fishes drink water and excrete only a small volume of very concentrated urine. In most fishes, the gills and gut are largely responsible for the excretion of surplus salts.

<u>The swim bladder</u>. This organ controls the fish's buoyancy and is used for hearing in some species. Most of the swim bladder is not permeable to gases, because it is poorly vascularised (has few blood vessels) and is lined with sheets of guanine crystals. A fish swimming in the water expends less energy if it is neutrally buoyant (that is, it neither sinks nor floats). If this fish starts to descend, the increased pressure from the water surrounding the fish results in a compression of the gas inside the swim bladder. The fish becomes negatively buoyant and will tend to sink. Conversely, if a fish swims into shallower water, there is a decrease in water pressure and so the gas in the swim bladder expands, and the fish tends to float upwards. The swim bladder helps to solve the problems associated with variations of pressure, and thus buoyancy. Not all fishes have a swim bladder. Sharks for example do not have a swim bladder, and many species such as the Grey Nurse Shark use a different strategy, which includes having a large oily liver and specialised body shape to maintain buoyancy.

<u>Gill filaments</u>. Just like the lungs of humans, gills of fishes are the sites where oxygen is absorbed and carbon dioxide is removed. In addition, the gills are responsible to a varying degree for regulation of the levels of various ions and the pH of the blood. The gill filaments of bony fishes (also known as a primary lamellae) are complex structures which have a large surface area. Off each are numerous smaller secondary lamellae. Tiny blood capillaries flow through the secondary lamellae of each gill filament. The direction of blood flow is opposite to that of water flow. This ensures that as the blood flows along each secondary lamella, the water flowing beside it always has a higher oxygen concentration than that in the blood. In this way oxygen is taken up along the entire length of the secondary lamellae. Active swimming fishes have well developed gill filaments to maximise the amount of oxygen that can be absorbed. Less active, bottom-dwelling fishes generally have much smaller gill filament volumes. Not all fishes rely totally on their gills to breathe. Some species, especially when they are young, absorb a large proportion of their oxygen requirements through the skin. Others species have well developed lungs for breathing air, and will in fact drown if they do not have access to the surface.

<u>Gill rakers.</u> These organs aid in the fish's feeding. The shape and number of gill rakers are a good indication of the diet of the fish. Fishes which eat large prey such as other fishes and molluscs have short, widely spaced gill rakers. This type of gill raker prevents the prey item from escaping between the gills. Fishes which eat smaller prey have longer, thinner and more numerous gill rakers. Species which feed on plankton and other tiny suspended matter have the longest, thinnest and most numerous gill rakers, with some species having over 150 on the lower arch alone.